Speech Emotion Recognition Using Deep Learning

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***Abstract*—This study explores the application of deep learning techniques in recognizing emotional states from spoken language. Specifically, we employ Convolutional Neural Networks (CNNs) and the HuBERT model to analyze the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). Our findings suggest that deep learning models, particularly the HuBERT model, exhibit significant potential in accurately identifying speech emotions. The models were trained and tested on a dataset containing various emotional expressions, including happiness, sadness, anger, and fear, among others. The experimentation involved preprocessing the audio data, feature extraction using Mel Frequency Cepstral Coefficients (MFCCs), and implementing deep learning architectures for emotion classification. The HuBERT model, with its advanced self-supervised learning mechanism, outperformed traditional CNNs in terms of accuracy and efficiency. This research highlights the importance of selecting appropriate deep learning models and feature sets for the task of speech emotion recognition. Our analysis demonstrates that the HuBERT model, by leveraging contextual information and temporal dynamics in speech, offers a promising approach for developing more sensitive and accurate SER systems. These systems have potential applications in various fields, including mental health assessment, interactive voice response systems, and educational software, by enabling machines to understand and respond to human emotions more effectively. The findings of this study contribute to the ongoing discussion in the field of artificial intelligence about the best practices for implementing deep learning techniques in speech processing tasks.**

Keywords—Speech Emotion Recognition, Deep Learning, Convolutional Neural Networks, HuBERT Model, RAVDESS Dataset

I. Introduction

The advancement of spoken language processing, intersecting with natural language processing, cognitive sciences, and human-machine interaction (HMI), has significantly propelled the development of adaptive and responsive human-machine interfaces. Speech Emotion Recognition (SER) emerges as a critical component in enhancing the naturalness and effectiveness of human-machine dialogue systems. Its relevance spans across smart environments, virtual assistants, and call centres, where emotional nuances in speech profoundly influence user interactions. This research delves into deploying deep learning techniques for SER, leveraging the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) for training and evaluating deep neural network (DNN) models. The study embarks on an extensive literature review on SER's recent methodologies and advancements. It endeavours to construct a comprehensive SER system pipeline, utilizing deep learning models for emotion detection from audio data. Utilizing Python and deep learning libraries such as TensorFlow and Keras, the research explores common DNN architectures including Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTMs), and Recurrent Neural Networks (RNNs) for SER. The project aims to implement a SER model, assessing its performance based on accuracy, precision, recall, and F1-score, and to fine-tune this model for optimized performance. Additionally, the effectiveness of pre-trained models, like HuBERT and ResNet, is compared against custom-built models. Finally, the paper outlines a potential business plan for the commercialization of SER technology, suggesting avenues for integrating emotional recognition capabilities into existing and future products and services. This contribution aims to advance HMI by enabling systems to more effectively understand and respond to human emotions.

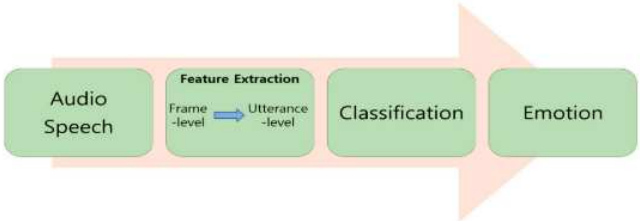

Fig. 1. Steps of SER

II. Speech Emotion Recognition systems

The field of artificial intelligence (AI) has catalysed significant advancements in human-machine communication, notably through the development of speech emotion recognition (SER). SER has gained prominence for its ability to discern nuanced emotional states from human speech, which is crucial for a variety of applications including entertainment, automotive safety, virtual assistants, healthcare, customer service centres, and e-learning platforms. The integration of SER into conversational systems like Google Assistant, Siri, and Alexa exemplifies its utility in enhancing user engagement and satisfaction by facilitating more natural and intuitive interactions between humans and machines.

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The pursuit of accurate emotion detection from speech presents complex challenges due to the variability of utterances and the subtlety of emotional expressions. Achieving high levels of SER performance necessitates navigating through intricate processes, including the pre-processing of audio data, feature extraction, and the classification of emotions. The endeavour to refine SER capabilities continues to be a methodological quest for algorithms that can surpass existing benchmarks, underscoring the intricacy and the multidimensional nature of human emotions as conveyed through speech.

III. Common Audio Features Extracted for Sound Classification Problems

In any machine learning project focused on audio, the initial step involves collecting audio signal data, which then must be transformed into features suitable for algorithmic processing. A key part of this process is defining and extracting the most crucial audio features for model construction, specifically for audio classification tasks.

A. *Mel Frequency Cepstral Coefficients (MFCCs)*

Utilizing the Librosa library in Python, one can extract significant features such as Mel Frequency Cepstral Coefficients (MFCCs). MFCCs are pivotal in capturing the timbral and textural qualities of sound within the frequency domain, closely approximating the human auditory system's response. These coefficients are influenced by the shape of the human vocal tract, including the tongue and teeth, playing a vital role in the precise representation of sounds. MFCCs effectively illustrate the nuanced differences in sound perception through the Mel scale, accounting for the way humans discern pitch differences across a broad frequency range from 20Hz to 20kHz. This ability to reflect the perceptual properties of sound makes MFCCs invaluable for audio classification models, enabling a more accurate and human-like processing of audio data.

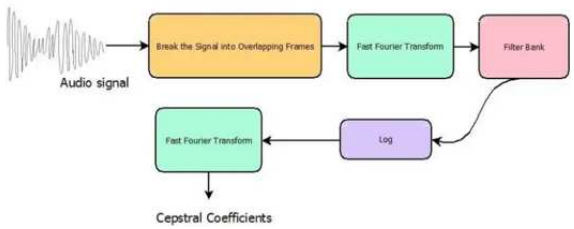

Fig. 2. Flowchart for obtaining MFCC coefficients

$$Mel(f) = 2595\log\left(1 + \frac{f}{100}\right) \quad (1)$$

B. *Short-Time Fourier Transform (STFT)*

The Short-time Fourier Transform (STFT) is a method that applies Fourier transforms to portions of a signal to obtain frequency information localized in time. Unlike the standard Fourier transform, which gives an average frequency overview for the whole signal, STFT can capture how frequency components change over time. This is achieved by dividing the signal into fixed-size frames (e.g., 2048 samples) and transforming each separately, making STFT essential for processing audio data in machine learning applications. The result is a spectrogram that maps time, frequency, and magnitude, providing a comprehensive representation of the signal. This spectrogram serves as the basis for extracting audio features, which are then structured into an array for neural network models to classify sounds.

Fig. 3. STFT of a sound signal

C. *Chroma*

Chroma feature, a quality of a pitch class which refers to the "color" of a musical pitch, which can be decomposed in into an octave-invariant value called "chroma" and a "pitch height" that indicates the octave the pitch is in.

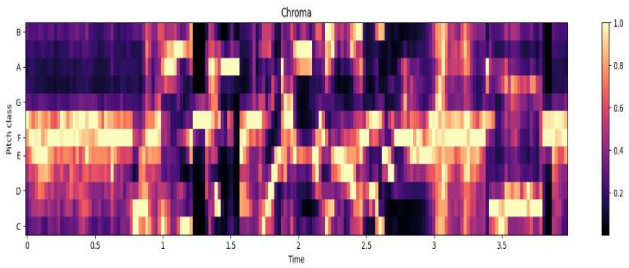

Fig. 4. Typical Chroma spectrogram of a sound signal

The audio features explained in this section therefore STFT, MFCC's and Chroma will be the features extracted from an audio signal and put into an array of features then fed into a model for emotion classification. This defines the Deep Learning pre-processing pipeline for audio data as shown in Fig. 5.

IV. The RAVDESS Dataset

Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) contains 1440 files: 60 trials per actor x 24 actors = 1440. The RAVDESS contains 24 professional actors (12 female, 12 male), vocalizing two lexically-matched statements in a neutral North American accent. Speech emotions includes calm, happy, sad, angry, fearful, surprise, and disgust expressions. Each expression is produced at two levels of emotional intensity (normal, strong), with an additional neutral expression.

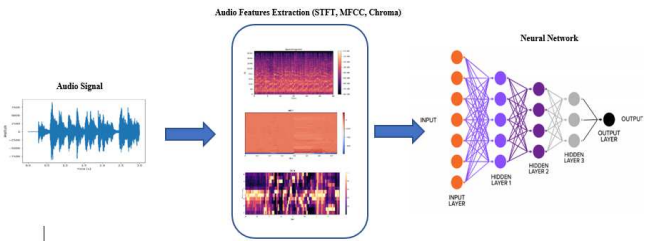

Fig. 5. Deep Learning pre-processing pipeline for Audio Data

V. Data Preparation

This section defines the steps under which the audio signal is pre-processed before being fed into the deep learning model for training, testing and then eventually emotion classification from speech. These steps can be defined as:

1. The sample audio is provided as input from RAVDESS dataset after importing the whole dataset into the script.

Due to the fact that this research work is intended to eventually be employed in a call-center i.e., only four

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emotions that are considered critical in such an industry i.e., Neutral, Sad, Angry and Happy are extracted from the RAVDESS dataset. Then they undergo signal processing before and training, testing and eventually being classified respectively.

2. A loop was made through RAVDESS directory to collect from audio folders the emotion, and gender of speaker and a plot of all collected audio files and emotions have been plotted.

3. Each of these audio files goes into feature extraction process as discussed in section III.

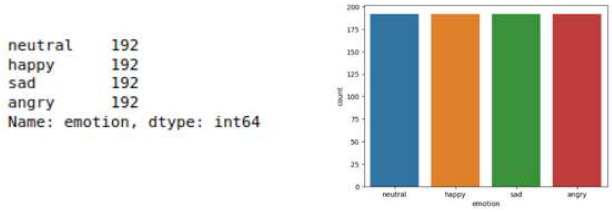

Fig. 6. Label Counts of intended emotions

4. All the extracted features of these emotions are then concatenated in a data frame which will be the input for the CNN model for emotion classification.

VI. CNN MODEL TRAINING AND TESTING

A conventional CNN architecture, as described in literature, designed for emotion classification. This model incorporates eight convolutional layers, all utilizing the ReLU activation function. To mitigate the risk of overfitting, dropout layers are strategically included in the model. The architecture begins with the first convolutional layer, which is initialized to match the dimensions of the x_train input variable, specifically (218,1). The culmination of the model is a dense layer with four units, reflecting the number of target emotion classes. The model employs the Adam optimizer, with a specified learning rate of 0.0001, to facilitate the learning process as summarized by the table below. Test split defined at 20% and training 80%.

TABLE I. CNN MODEL ARCHITECTURE

Layer	*Filters/Units*	*Kernel/Pool Size*	*Activation*	*Notes*
Conv1D	256	8	ReLU	Input shape: (218, 1). 2 layers
BatchNormalization	-	-	-	-
Dropout	-	-	-	0.5, after Conv and before MaxPooling
MaxPooling1D	-	8	-	-
Conv1D	128	8	ReLU	4 layers with Dropout after every layer
BatchNormalization	-	-	-	-
MaxPooling1D	-	8	-	-
Conv1D	64	8	ReLU	2 layers
Flatten	-	-	-	-
Dense	4	-	Softmax	Target class: emotions
Optimizer	-	-	-	Adam, Learning rate: 0.0001

Model was then compiled as follows and a test accuracy of 40% has been achieved. The loss was chosen as 'categorical cross entropy', epochs= 80, batch size=32. However, this accuracy is considered very low for such a task and such a dataset. This accuracy was the maximum reached by this model even after parameters tuning. Hence, adoption of another model to achieve a higher accuracy given the same inputs was considered. The model under consideration was the HuBERT model which will be discussed in the following section.

VII. HUBERT MODEL TRAINING AND TESTING

The HuBERT model, built on the Transformer architecture, is specifically designed for tasks like speech emotion detection. It uniquely identifies discrete elements in speech, enabling a deeper understanding of emotional cues. The training of HuBERT for emotion detection on the RAVDESS dataset, which includes emotions such as angry, sad, happy, and neutral, consists of an initial unsupervised pre-training phase. Here, the model learns from unlabelled data by predicting masked speech segments, developing a generic grasp of speech patterns. Subsequently, it undergoes fine-tuning with labelled data from RAVDESS, focusing on the four specified emotions. This process refines the model's predictive capabilities. Testing then validates its performance on unseen data, assessing its precision in emotion identification. HuBERT's advanced architecture and training methodology offer significant potential for enhanced accuracy in speech emotion recognition tasks. And as a result, this is achieved. The HuBERT model resulted in an accuracy of 83.77% as compared to 40% on the CNN model.

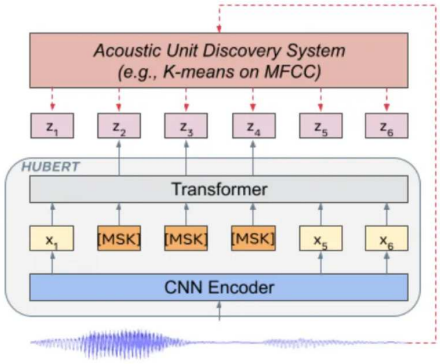

Fig. 7. HuBERT Model Architecture

VIII. RESULTS AND DISCUSSIONS

This chapter aims to detail the performance outcomes of two Speech Emotion Recognition (SER) models: a custom CNN model and the HuBERT base model, utilizing the amended RAVDESS dataset. Performance evaluation involves confusion matrices, various evaluation metrics, and training accuracy, focusing on the models' capability to discern four specific emotions: neutral, happy, sad, and angry, following label balancing. The CNN model, crafted with convolutional, pooling, and dropout layers, underwent training and testing on the cleaned dataset, assessing its proficiency in emotion identification. Conversely, the pre-trained HuBERT model received fine-tuning on the same dataset, with its performance similarly evaluated. Notably, the HuBERT base model significantly surpassed the CNN model, achieving an 83.77% overall accuracy compared to the CNN's 40.97%. This chapter will present a detailed comparison between the models, showcasing their respective strengths and weaknesses in SER through confusion matrices and key evaluation metrics like precision, recall, and F1 score, offering a thorough analysis of each model's effectiveness in emotion recognition.

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A. Confusion Matrix

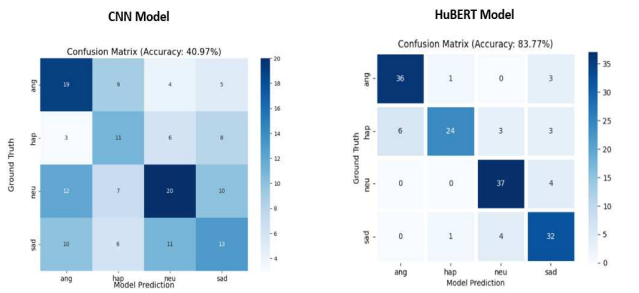

Fig. 8. Confusion matrix plots of CNN and HuBERT Model

Examining the confusion matrices, as shown in Fig. 8, reveals a clear disparity in performance between two models, particularly evident in the diagonal segments which denote accurate emotion classifications. The use of colour spectrograms, with darker and lighter shades of blue indicating higher and lower accuracy respectively, visually suggests the HuBERT model's superior ability in emotion recognition. However, this observation, primarily based on correct classifications of samples, does not fully encapsulate overall performance. Thus, to attain a more comprehensive analysis, additional evaluation metrics were employed. These metrics, discussed in the previous section, allow for a deeper examination and comparison of the models, ultimately leading to a more grounded determination of which model outperforms the other.

B. Model Predictions on Test Dataset

The accuracy of the model on test dataset is determined by

$$\frac{\textit{Total Correct Predictions}}{\textit{Total number of predicitons}} x\ 100 \tag{2}$$

```
Total Correct Predictions: 63
Total Incorrect Predictions: 91

     File Ground Truth Model Prediction  Correct
0       1          neu             hap    False
1       2          hap             hap     True
2       3          sad             neu    False
3       4          sad             hap    False
4       5          hap             neu    False
..    ...          ...             ...      ...
149   150          neu             ang    False
150   151          neu             ang    False
151   152          sad             neu    False
152   153          neu             neu     True
153   154          ang             sad    False

[154 rows x 4 columns]
```

CNN model prediction on test dataset

```
Total Correct Predictions: 129
Total Incorrect Predictions: 25

     File Ground Truth Model Prediction  Correct
0       1          sad             neu    False
1       2          neu             ang    False
2       3          neu             ang    False
3       4          neu             hap    False
4       5          hap             hap     True
..    ...          ...             ...      ...
149   150          neu             hap    False
150   151          ang             ang     True
151   152          hap             hap     True
152   153          hap             neu    False
153   154          hap             ang    False

[154 rows x 4 columns]
```

HuBERT model prediction on test dataset

Fig. 9. Model prediction performance of CNN and HuBERT on test dataset

Fig. 9, illustrates that the HuBERT model significantly surpasses the CNN model in terms of accuracy on the test dataset, achieving 129 correct predictions out of 154 total cases. In contrast, the CNN model managed only 63 correct predictions, highlighting the superior performance of the HuBERT model.

- CNN model accuracy on test dataset:

$$\frac{\textit{Total Correct Predictions}}{\textit{Total number of predicitons}} x\ 100 = \frac{63}{154} x100 = 40.97\%$$

- HuBERT model accuracy on test dataset:

$$\frac{\textit{Total Correct Predictions}}{\textit{Total number of predicitons}} x\ 100 = \frac{129}{154} x100 = 83.76\%$$

C. Precision, Recall, F1-score:

	precision	recall	f1-score
ang	0.25	0.27	0.26
hap	0.20	0.20	0.20
neu	0.22	0.20	0.21
sad	0.28	0.30	0.29

CNN model evaluation metrics

Emotion	Precision	Recall	F1-Score
ang	0.230769	0.300000	0.260870
hap	0.296296	0.355556	0.323232
neu	0.428571	0.394737	0.410959
sad	0.461538	0.292683	0.358209

HuBERT model evaluation metrics

Fig. 10. Evaluation metrics on the 4 emotions for CNN and HuBERT model

Fig. 10, shows that, based on evaluation metrics, the HuBERT model significantly outperforms the CNN model, achieving higher scores in all metrics, particularly noting a substantial increase in precision and F1-score for the "neutral" and "sad" emotions compared to the CNN model.

IX. Conclusion

This research successfully implemented and compared two Speech Emotion Recognition (SER) models, a customized CNN and the HuBERT base model, on the RAVDESS dataset focusing on four primary emotions: neutral, sad, angry, and happy. Performance analysis revealed the HuBERT model significantly outperformed the CNN model, achieving an 83.77% accuracy versus 40.97%, thus becoming the preferred choice for optimal SER performance.

Key outcomes:

- Development of a comprehensive SER system pipeline, encompassing audio data collection through to emotion recognition.
- Utilization of Python and deep learning libraries (TensorFlow, Keras, PyTorch, Librosa) to build the SER framework.
- Exploration of common neural network models (CNN, LSTM, RNN) and the implementation of a customized CNN model.
- Investigation and adoption of pre-trained models like HuBERT and ResNet, with HuBERT being chosen for its superior performance.
- Fine-tuning of both models to enhance performance, emphasizing the importance of hyperparameter optimization.

Future research directions are proposed to extend the work further, including:

1. Expanding training data beyond the RAVDESS dataset. Aiming to recognize all eight emotions provided in the RAVDESS dataset.
2. Exploring and comparing additional models such as ResNet.
3. Developing a real-time SER system for potential commercial deployment.

This study advances SER research, offering a foundation for future exploration. The proposed system and findings have broad potential applications across industries, from healthcare to entertainment, highlighting the significance of advancing SER technology for real-world applications.

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